

SHIVANSH THAKUR

AI Engineer: LLM Applications, RAG, AI Agents & FastAPI Services

+91 70182 13168 · shivanshthakur1708@gmail.com

linkedin.com/in/shivansh-thakur1708 · github.com/shivu0070 · shivu0070.github.io/portfolio

SUMMARY

AI engineer with ~1 year as one of two AI/ML engineers at a startup, independently building LLM applications end to end: RAG pipelines (embeddings, semantic search, vector database), LLM API integration (Gemini, OpenAI), structured outputs, multi-agent workflows with tool use, and FastAPI + PostgreSQL services, backed by pytest suites and measured accuracy. Includes offline local-LLM inference on CPU-constrained hardware.

WORK EXPERIENCE

Jr. Data Scientist, AI Engineering

Nov 2025 – Present

Harij Softech Solutions Pvt. Ltd. · AI & enterprise-software startup · On-site, India

- Designed and developed a local-LLM RAG chatbot (Ollama + FAISS vector search over document embeddings) that lets employees query company policies, grounding answers in retrieved policy text with an "ask HR" fallback as a guardrail against unsupported answers.
- Client project, ConsultOpen (consultopen.com): designed and built, using the Gemini API, an LLM-based ATS resume scorer (100-point rubric, structured JSON output) and FastAPI + PostgreSQL services recommending therapists and exercises from mood check-ins.
- Designed and built from scratch a layout-independent invoice extractor (LightOnOCR-2 vision model, then llama3.2 producing structured JSON) with no per-layout code, replacing a 51-function layout-repair pipeline: 92.9% of 424 hand-verified cells correct; 98.6% of values matched on 24 unseen invoices; runs on a 4 GB GPU.
- Architected and developed end to end an offline medical-PDF splitter using a vision-language model (Qwen2.5-VL-7B) to assign pages to patients: 97.2% of 281 pages routed correctly across 18 test documents; all 40 patients identified in a separate 240-page batch; 277 pytest tests.
- Built OCR extractors for dispatch notes and vehicle RC certificates (PaddleOCR, ~10 s/page on CPU), automating data entry and routing low-confidence fields to manual review.
- Built from scratch a real-time ALPR pipeline (YOLOv8 + SORT + PaddleOCR) with character voting across up to 12 reads per vehicle, and trained its YOLOv8s plate detector on 3,715 images (97.5% precision, 98.7% mAP@50).

Data Scientist Intern

Jan 2025 – Oct 2025

Analytics Career Connect · Remote

- Analysed delivery performance across 180K+ orders in the public DataCo Supply Chain dataset (Python, SQL/SQLite, Power BI), surfacing that ~54% of deliveries ran late and which shipping modes and categories drove profit.
- Identified high-value loyalty cohorts for marketing targeting through RFM and CLTV segmentation of 4,300+ customers (UCI Online Retail), with Power BI dashboards cross-validated against SQL.

TECHNICAL SKILLS

LLMs & GenAI: LLM APIs (Google Gemini, OpenAI), Anthropic Claude models (via Claude Code), Ollama and local LLMs, RAG architecture, embeddings, semantic search, vector databases (FAISS), prompt engineering, structured outputs (JSON, Pydantic), AI agents and multi-agent workflows, tool use, LLM evaluation, guardrails and fallbacks, Whisper

Backend & Data: FastAPI, API-based architectures, Docker, PostgreSQL, SQLAlchemy, SQL, Pydantic, pytest, Git, Linux, debugging and troubleshooting

AI / ML: Python, PyTorch, HuggingFace Transformers, scikit-learn

Computer Vision & OCR: YOLOv8, OpenCV, PaddleOCR, Tesseract, vision-language models

PROJECTS

Autonomous Dev Team

github.com/shivu0070/autonomous-dev-team

Designed and built a multi-agent app-delivery pipeline orchestrating Claude Code subagents (PM, architects, developers) with an independent Playwright + vision QA agent; 12 apps built, all passing acceptance criteria written by a separate PM agent.

Local Voice AI Agent

github.com/shivu0070/local-voice-ai

Built from scratch a fully local voice assistant with sub-1s barge-in (Whisper, Silero VAD, Ollama, Piper TTS over LiveKit WebRTC), streaming TTS and appointment booking; evaluated with 300+ scripted queries across 4 local LLMs.

WhatsApp Intent Router

github.com/shivu0070/whatsapp-intent-router

Designed and fine-tuned a 5-intent DistilBERT router for Hinglish WhatsApp messages; resolved overlapping labels in an earlier 8-intent design and passed 36/36 hand-written smoke tests.

EDUCATION

MCA, AI & Machine Learning · Chandigarh University, 2023 – 2025 | **BCA** · Himachal Pradesh University, 2020 – 2023